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Connector

Abstract

A connector includes a housing having a first connection portion and a second connection portion adjacent to the first connection portion, a first cap and a second cap pivotally supported by the housing respectively to be displaceable between an open position where the first and second connection portions are exposed to outside and a closed position where the first and second connection portions are shielded from the outside, a first lock and a second lock rotatably supported by the housing to lock the first cap and the second cap respectively, and a power transmission member rotatably supported by the housing. The power transmission member is configured to transmit a rotational force in a lock releasing direction of the second lock to the first lock such that the first cap and the second cap are integrally displaced to the open position.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application is a continuation application of PCT/JP2022/040844 that claims priority to Japanese Patent Application No. 2021-179593 filed on Nov. 2, 2021, the entire content of which is incorporated herein by reference.

TECHNICAL FIELD

(1) The present disclosure relates to a connector.

BACKGROUND ART

(2) An electric vehicle runs by charging a built-in power supply (rechargeable battery) mounted therein and driving a DC motor. As a method of charging a built-in power supply, there are normal charging (so-called AC charging) performed using a commercial power supply at home, and quick charging (so-called DC charging) performed by a power feeding device installed in a power feeding station or the like. For this reason, a power receiving connector provided with a power receiving unit for normal charging (hereinafter also referred to as a “normal power receiving unit”) and a power receiving unit for quick charging (hereinafter also referred to as a “quick power receiving unit”) is mounted on the electric vehicle. Such a power receiving connector is provided with caps for shielding the normal power receiving unit and the quick power receiving unit from being exposed to the outside at the time of non-charging as disclosed, for example, in JPH08-31497A.

(3) An alternating current is used in the normal charging, and a direct current is used in the quick charging.

(4) In general, standards are established for power receiving connectors, and one of the standards is to use only the normal power receiving unit during normal charging, and to use the normal power receiving unit in addition to the quick power receiving unit during quick charging. In the above standard, during normal charging, only a cap for the normal power receiving unit is required to be opened, but during quick charging, the cap for the normal power receiving unit is also required to be opened in addition to a cap for the quick power receiving unit.

(5) A connector according to the embodiment is excellent in workability in which two caps can be opened simultaneously.

SUMMARY OF INVENTION

(6) A connector includes a housing having a first connection portion and a second connection portion adjacent to the first connection portion, a first cap pivotally supported by the housing to be displaceable between an open position where the first connection portion is exposed to outside and a closed position where the first connection portion is shielded from the outside, a second cap pivotally supported by the housing to be displaceable between an open position where the second connection portion is exposed to the outside and a closed position where the second connection portion is shielded from the outside, a first lock rotatably supported by the housing to lock the first cap when the first cap is located at the closed position, a second lock rotatably supported by the housing to lock the second cap when the second cap is located at the closed position, and a power transmission member rotatably supported by the housing. The power transmission member is configured to transmit a rotational force in a lock releasing direction of the second lock to the first lock as a rotational force in a lock releasing direction of the first lock such that the first cap and the second cap are integrally displaced to the open position.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a perspective view of a connector according to an embodiment of the present disclosure, showing a non-charging aspect;

(2) FIG. 2 is a view showing an aspect of the connector shown in FIG. 1 during normal charging;

(3) FIG. 3 is a view showing an aspect of the connector shown in FIG. 1 during quick charging;

(4) FIG. 4 is a front view of the connector shown in FIG. 1;

(5) FIG. 5 is an exploded perspective view of the connector shown in FIG. 1;

(6) FIG. 6 is a front view of a power transmission member in the aspect shown in FIG. 1; and

(7) FIG. 7 is a front view of the power transmission member in the aspect shown in FIG. 3.

DESCRIPTION OF EMBODIMENTS

Embodiment

(8) Hereinafter, a connector **1** according to an embodiment of the present disclosure will be described with reference to the drawings. The connector **1** is, for example, a power receiving connector (so-called vehicle inlet) mounted on a vehicle of an electric vehicle.

(9) Hereinafter, for convenience of description, “front”, “rear”, “left”, “right”, “upper”, and “lower” are defined as illustrated in FIGS. **1** to **7**. A “front-rear direction”, a “left-right direction”, and an “upper-lower direction” are orthogonal to one another. When the connector **1** is mounted on a vehicle, the “front-rear direction”, the “left-right direction”, and the “upper-lower direction” correspond to the front-rear direction, the left-right direction, and the upper-lower direction of the vehicle, respectively. In addition, the front-rear direction corresponds to a fitting direction of the connector **1** and a power feeding connector. A locking direction and a lock releasing direction in the present embodiment are respectively the locking direction and the lock releasing direction of a first lock **40** and a second lock **50**.

(10) As shown in FIGS. **1** to **5**, the connector **1** includes a housing **10** having a first connection portion **13** and a second connection portion **14**, a first cap **20** rotatably supported by the housing **10**, a second cap **30** rotatably supported by the housing **10**, a first lock **40** capable of locking the first cap **20**, a second lock **50** capable of locking the second cap **30**, and a power transmission member **60** rotatably supported by the housing **10**. Hereinafter, the housing **10**, the first cap **20**, the second cap **30**, the first lock **40**, the second lock **50**, and the power transmission member **60** constituting the connector **1** will be described in this order.

(11) First, the housing **10** will be described. As shown in FIGS. **1** to **5**, the housing **10** is integrally formed with a substantially rectangular box-shaped body portion **11**, and a hood portion **12** which is provided on a front end surface of the body portion **11**, opens forward, and has an accommodation space **S** inside.

(12) A pair of cap-side support shaft portions **15** protruding leftward are provided on an outer surface of a left peripheral wall of the hood portion **12**. The pair of cap-side support shaft portions **15** are arranged side by side at a predetermined interval in the upper-lower direction on the left peripheral wall of the hood portion **12**.

(13) A pin-shaped cap rotation shaft portion **15a** is provided on each of the pair of cap-side support shaft portions **15**. A pair of arm portions **22** of the first cap **20**, which will be described later, are rotatably supported on the upper cap rotation shaft portion **15a**, and a pair of arm portions **32** of the second cap **30**, which will be described later, are rotatably supported on the lower cap rotation shaft portion **15a**. In addition, a spring **80** is pivotally supported on each of the cap rotation shaft portions **15a** (see FIGS. **2** and **3**).

(14) A pair of lock-side support shaft portions **16** protruding rightward are provided on an outer surface of a right peripheral wall of the hood portion **12**. The pair of lock-side support shaft portions **16** are arranged side by side at a predetermined interval in the upper-lower direction on the right peripheral wall of the hood portion **12**.

(15) The lock-side support shaft portion **16** is provided with a pin-shaped lock rotation shaft portion **16a**. The lock rotation shaft portion **16a** is one rotation shaft portion supported by the pair of lock-side support shaft portions **16** (see FIG. **5**). The lock rotation shaft portion **16a** rotatably supports the first lock **40** to be described later at a position corresponding to the upper lock-side support shaft portion **16**, rotatably supports the second lock **50** to be described later at a position corresponding to the lower lock-side support shaft portion **16**, and rotatably supports the power transmission member **60** between the pair of lock-side support shaft portions **16**. In addition, springs **70** are pivotally supported at positions of the lock rotation shaft portion **16a** corresponding to the respective lock-side support shaft portions **16** (see FIG. **5**).

(16) The housing **10** further includes the first connection portion **13** and the second connection portion **14** in the accommodation space **S** of the hood portion **12** (see FIG. **3**). A plurality of terminals (not shown) are accommodated in the first connection portion **13**. Similarly, a plurality of

terminals (not shown) are accommodated in the second connection portion **14**. In the connector **1**, a power feeding connector for normal charging is connected to the first connection portion **13** during normal charging, and a connector for quick charging is connected to the first connection portion **13** and the second connection portion **14** during quick charging.

(17) Next, the first cap **20** will be described. As shown in FIGS. **1** to **4**, the first cap **20** is integrally formed with a body portion **21** that closes an upper portion of an opening of the hood portion **12** at a closed position described later, the pair of arm portions **22** that are rotatably supported by the cap rotation shaft portion **15a** of the housing **10**, and a locking portion **23** that is locked to the first lock **40** at the closed position. The body portion **21** has a shape corresponding to the opening of the hood portion **12**.

(18) The first cap **20** can be displaced between an open position (see FIGS. **2** and **3**) where the first connection portion **13** of the housing **10** is exposed to the outside, and the closed position (see FIGS. **1** and **4**) where the first connection portion **13** is shielded from the outside (that is, the upper portion of the opening of the hood portion **12** is closed).

(19) Further, when the first cap **20** is located at the closed position, a force in an opening direction of the first cap **20** (that is, the clockwise direction when viewed from above in an axial direction) is applied by the spring **80** which is axially supported by the cap rotation shaft portion **15a**. That is, when the first cap **20** is not locked to the first lock **40**, the first cap **20** is maintained in the open position by the spring **80**. Note that when the first cap **20** is located at the open position, it is sufficient that the force in the opening direction of the first cap **20** is applied by the spring **80** to an extent that the first cap **20** is not displaced to the closed position by itself.

(20) Next, the second cap **30** will be described. As shown in FIGS. **1** to **4**, the second cap **30** is integrally formed with a body portion **31** that closes a lower portion of the opening of the hood portion **12** at a closed position described later, the pair of arm portions **32** that are rotatably supported by the cap rotation shaft portion **15a** of the housing **10**, and a locking portion **33** that is locked to the second lock **50** at the closed position. The body portion **31** has a shape corresponding to the opening of the hood portion **12**.

(21) The second cap **30** can be displaced between an open position (see FIG. **3**) where the second connection portion **14** of the housing **10** is exposed to the outside, and the closed position (see FIGS. **1** and **4**) where the second connection portion **14** is shielded from the outside (that is, the lower portion of the opening of the hood portion **12** is closed).

(22) That is, when the first cap **20** and the second cap **30** are both located at the closed position, the opening of the hood portion **12** is closed by the first cap **20** and the second cap **30**.

(23) Further, when the second cap **30** is located at the closed position, a force in an opening direction of the second cap **30** (that is, the clockwise direction when viewed from above in the axial direction) is applied by the spring **80** which is axially supported by the cap rotation shaft portion **15a**. That is, when the second cap **30** is not locked to the second lock **50**, the second cap **30** is maintained in the open position by the spring **80**. Note that when the second cap **30** is located at the open position, it is sufficient that the force in the opening direction of the second cap **30** is applied by the spring **80** to an extent that the second cap **30** is not displaced to the closed position by itself.

(24) Next, the first lock **40** will be described. As shown in FIG. **5**, the first lock **40** is integrally formed with a body portion **41** having a locked portion (not shown) to which the locking portion **23** of the first cap **20** is locked when the first cap **20** is located at the closed position, and a pair of arm portions **42** rotatably supported by the lock rotation shaft portion **16a** at a position corresponding to the upper lock-side support shaft portion **16**. The locked portion is, for example, a recessed portion (in FIG. **1**, a portion of a rear end surface of the body portion **41** is recessed forward) accommodating the locking portion **23** when the first cap **20** is located at the closed position, and the locking portion **23** is locked by accommodating the locking portion **23**.

(25) A position where the first lock **40** locks the first cap **20** is set as an initial position. In addition, a force in the locking direction (that is, the clockwise direction when viewed from above in the

axial direction) is applied to the first lock **40** by the spring **70** which is rotatably supported by the lock rotation shaft portion **16a**. That is, the first lock **40** is maintained at the initial position unless operated in the lock releasing direction (that is, the counterclockwise direction when viewed from above in the axial direction), and even if the first lock **40** is operated in the lock releasing direction, the spring **70** returns the first lock **40** to the initial position after the operation is completed.

(26) Next, the second lock **50** will be described. As shown in FIG. 5, the second lock **50** is integrally formed with a body portion **51** having a locked portion (not shown) to which the locking portion **33** of the second cap **30** is locked when the second cap **30** is located at the closed position, and a pair of arm portions **52** rotatably supported by the lock rotation shaft portion **16a** at a position corresponding to the lower lock-side support shaft portion **16**. The locked portion is, for example, a recessed portion (in FIG. 1, a portion of a rear end surface of the body portion **51** is recessed forward) accommodating the locking portion **33** when the second cap **30** is located at the closed position, and the locking portion **33** is locked by accommodating the locking portion **33**.

(27) A position where the second lock **50** locks the second cap **30** is set as an initial position. In addition, a force in the locking direction (that is, the clockwise direction when viewed from above in the axial direction) is applied to the second lock **50** by the spring **70** which is rotatably supported by the lock rotation shaft portion **16a**. That is, the second lock **50** is maintained at the initial position unless operated in the lock releasing direction (that is, the counterclockwise direction when viewed from above in the axial direction), and even if the second lock **50** is operated in the lock releasing direction, the spring **70** returns the second lock **50** to the initial position after the operation is completed.

(28) Next, the power transmission member **60** will be described. As shown in FIGS. 6 to 7, the power transmission member **60** is integrally formed with a substantially columnar body portion **61** rotatably supported by the lock rotation shaft portion **16a**, a first extending portion **62** extending from an upper end portion of the body portion **61** in a predetermined direction, and a second extending portion **64** extending from a lower end portion of the body portion **61** in a direction different from the first extending portion **62**. Note that, taking FIG. 6 as an example, the first extending portion **62** extends toward a left front side, and the second extending portion **64** extends toward a right front side.

(29) A first contact portion **63** to come into contact with the first lock **40** only on a locking direction side (the left side of the paper surface in FIG. 4) protrudes upward from the first extending portion **62**. A second contact portion **65** to come into contact with the second lock **50** only on a lock releasing direction side (the right side of the paper surface in FIG. 4) protrudes downward from the second extending portion **64**. That is, an end surface (right end surface in FIG. 4) on a lock releasing direction side of the first contact portion **63** comes into contact with an end surface (left end surface in FIG. 4) on the locking direction side of the first lock **40**, and an end surface (left end surface in FIG. 4) on the locking direction side of the second contact portion **65** comes into contact with an end surface (right end surface in FIG. 4) on the lock releasing direction side of the second lock **50**. Therefore, the extending directions of the first extending portion **62** and the second extending portion **64** are set such that the first contact portion **63** comes into contact with the first lock **40** only on the locking direction side and the second contact portion **65** comes into contact with the second lock **50** only on the lock releasing direction side.

(30) The power transmission member **60** is disposed between the pair of lock-side support shaft portions **16**, and has a function of transmitting a rotational force in the lock releasing direction of the second lock **50** to the first lock **40** as a rotational force in the lock releasing direction of the first lock **40** so that the first cap **20** and the second cap **30** are integrally displaced to the open position during quick charging.

(31) The housing **10**, the first cap **20**, the second cap **30**, the first lock **40**, the second lock **50**, and the power transmission member **60** constituting the connector **1** have been described above.

(32) Next, each aspect of the connector **1** during non-charging, normal charging, and quick

charging will be described. As shown in FIGS. 1 and 4, at the time of non-charging of the electric vehicle, the opening of the hood portion 12 is closed by the first cap 20 and the second cap 30 in the connector 1 so that the first connection portion 13 and the second connection portion 14 of the housing 10 are not exposed to the outside.

(33) At this time, the first cap 20 is locked to the first lock 40, and the second cap 30 is locked to the second lock 50. Specifically, the locking portion 23 of the first cap 20 is locked to the locked portion (not shown) of the first lock 40, and the locking portion 33 of the second cap 30 is locked to the locked portion (not shown) of the second lock 50.

(34) For example, when the electric vehicle is charged by the normal charging, it is necessary to perform an opening operation on the first cap 20. Specifically, the first lock 40 is rotated in the lock releasing direction to release the locking between the locking portion 23 of the first cap 20 and the locked portion of the first lock 40. When the locking between the first cap 20 and the first lock 40 is released, the first cap 20 is returned to the open position by the spring 80. When a power feeding connector for normal charging is connected to the first connection portion 13, normal charging of the electric vehicle is started.

(35) That is, as shown in FIG. 2, at the time of normal charging of the electric vehicle, in the connector 1, the first cap 20 is positioned at the open position so that the first connection portion 13 is exposed to the outside, and the lower portion of the opening of the hood portion 12 is closed by the second cap 30 so that the second connection portion 14 is not exposed to the outside.

(36) After the operation in the lock releasing direction is completed, the first lock 40 is returned to the initial position by the spring 70. In FIG. 2, the first lock 40 is not returned to the initial position for convenience of description.

(37) Further, for example, when the electric vehicle is charged by quick charging, it is necessary to perform the opening operation on the first cap 20 and the second cap 30. Specifically, the second lock 50 is rotated in the lock releasing direction to release the locking between the locking portion 33 of the second cap 30 and the locked portion of the second lock 50, and to release the locking between the locking portion 23 of the first cap 20 and the locked portion of the first lock 40.

(38) As described above, in the power transmission member 60, the first contact portion 63 is in contact with the first lock 40 only in the locking direction side, and the second contact portion 65 is in contact with the second lock 50 only in the lock releasing direction side (see FIG. 4). Taking the aspect during non-charging shown in FIGS. 1 and 4 as an example, the right end surface of the first contact portion 63 is in contact with the left end surface of the first lock 40, and the left end surface of the second contact portion 65 is in contact with the right end surface of the second lock 50.

(39) Thus, when the second lock 50 is operated in the lock releasing direction, the rotational force is also transmitted to the second contact portion 65 in the lock releasing direction. That is, since the power transmission member 60 also rotates in the lock releasing direction, the first contact portion 63 naturally rotates in the lock releasing direction (see also arrows in FIG. 6).

(40) Since the right end surface of the first contact portion 63 and the left end surface of the first lock 40 are in contact with each other, the rotational force in the lock releasing direction of the first contact portion 63 is transmitted to the first lock 40. That is, the first lock 40 also rotates in the lock releasing direction. Accordingly, the locking between the second cap 30 and the second lock 50 is released, and the locking between the first cap 20 and the first lock 40 is released. In other words, in the connector 1, the first cap 20 and the second cap 30 are integrally displaced to the open position where the first connection portion 13 and the second connection portion 14 are exposed to the outside by the power transmission member 60.

(41) As described above, if the locking between the second cap 30 and the second lock 50 is released and the locking between the first cap 20 and the first lock 40 is released, the first cap 20 and the second cap 30 are returned to the open position by the respective springs 80. When the power feeding connector for quick charging is connected to the first connection portion 13 and the second connection portion 14, quick charging of the electric vehicle is started.

(42) That is, as shown in FIG. 3, at the time of quick charging of the electric vehicle, in the connector **1**, the first cap **20** is positioned at the open position so that the first connection portion **13** is exposed to the outside, and the second cap **30** is positioned at the open position so that the second connection portion **14** is exposed to the outside. In other words, at the time of quick charging of the electric vehicle, the connector **1** is in a state in which the opening of the hood portion **12** is not closed.

(43) After the operation in the lock releasing direction is completed, the first lock **40** and the second lock **50** are returned to the initial position by the respective springs **70**. In FIG. 2, the first lock **40** is not returned to the initial position for convenience of description.

(44) The first lock **40** and the second lock **50** according to the present embodiment have a so-called right opening structure. Therefore, in the present embodiment, rotation in the locking direction is clockwise rotation when viewed from above in the axial direction, and rotation in the lock releasing direction is counterclockwise rotation when viewed from above in the axial direction. In addition, the first cap **20** and the second cap **30** according to the present embodiment have a so-called left opening structure. Therefore, in the present embodiment, the opening direction of the first cap **20** and the second cap **30** corresponds to the clockwise direction when viewed from above in the axial direction. A closing direction of the first cap **20** and the second cap **30** corresponds to the counterclockwise direction when viewed from above in the axial direction.

(45) The respective aspects of the connector **1** during non-charging, normal charging, and quick charging have been described above.

(46) Note that the connector **1** according to the present embodiment conforms to the standard combined charging system type 2 (CCS 2). The connector **1** according to the present embodiment is also compatible with the standard combined charging system type 1 (CCS 1).

(47) According to the connector **1** of the present embodiment, the power transmission member **60** transmits the rotational force in the lock releasing direction of the second lock **50** that locks the second cap **30** to the first lock **40** that locks the first cap **20** as the rotational force in the lock releasing direction of the first lock **40**, so that the first cap **20** and the second cap **30** are integrally displaced to the open position where the first connection portion **13** and the second connection portion **14** are exposed to the outside. Thus, the connector **1** according to the present embodiment can simultaneously open the two caps of the first cap **20** and the second cap **30** at the time of quick charging.

(48) Further, according to the connector **1** of the present embodiment, the power transmission member **60** includes the first contact portion **63** that comes into contact with the first lock **40** only on the locking direction side, and the second contact portion **65** that comes into contact with the second lock **50** only on the lock releasing direction side. Thus, when the second lock **50** is operated in the lock releasing direction, the rotational force is also transmitted to the second contact portion **65** in the lock releasing direction, and the power transmission member **60** also rotates in the lock releasing direction. The rotational force is also transmitted to the first lock **40** via the first contact portion **63**, and the first lock **40** also rotates in the lock releasing direction. Thus, the connector **1** according to the present embodiment can simultaneously open the two caps of the first cap **20** and the second cap **30** at the time of quick charging.

(49) Further, according to the connector **1** of the present embodiment, the power transmission member **60** is supported by a rotation shaft common to the first lock **40** and the second lock **50**. Thus, the connector **1** according to the present embodiment can simultaneously open the two caps of the first cap **20** and the second cap **30** at the time of quick charging while reducing the number of components.

(50) Thus, the connector **1** according to the present embodiment is excellent in workability compared with a connector in the related art.

OTHER EMBODIMENTS

(51) The present disclosure is not limited to the above-described embodiment, and can be

appropriately modified, improved, or the like. In addition, materials, shapes, sizes, numbers, arrangement positions and the like of components in the above-described embodiments are freely selected and are not limited as long as the present disclosure can be implemented.

(52) Here, characteristics of the connector according to the present embodiment described above will be briefly summarized and listed in the following (i) to (iii). (i) A connector (1) including: a housing (10) having a first connection portion (13) and a second connection portion (14) adjacent to the first connection portion; a first cap (20) pivotally supported by the housing (10) to be displaceable between an open position (see FIGS. 2 and 3) where the first connection portion is exposed to outside and a closed position (see FIGS. 1 and 4) where the first connection portion is shielded from the outside; a second cap (30) pivotally supported by the housing (10) to be displaceable between an open position (see FIG. 3) where the second connection portion is exposed to the outside and a closed position (see FIGS. 1 and 4) where the second connection portion is shielded from the outside; a first lock (40) rotatably supported by the housing (10) to lock the first cap (20) when the first cap (20) is located at the closed position; a second lock (50) rotatably supported by the housing (10) to lock the second cap (30) when the second cap (30) is located at the closed position; and a power transmission member (60) rotatably supported by the housing (10), in which the power transmission member (60) is configured to transmit a rotational force in a lock releasing direction of the second lock (50) to the first lock (40) as a rotational force in a lock releasing direction of the first lock (40) such that the first cap (20) and the second cap (30) are integrally displaced to the open position. (ii) In the connector (1) according to (i), the power transmission member (60) includes a first contact portion (63) to come into contact with the first lock (40) only on a locking direction side, and a second contact portion (65) to come into contact with the second lock (50) only on a lock releasing direction side. (iii) In the connector (1) according to (i), the power transmission member (60) is supported by a rotation shaft (lock rotation shaft portion 16a) common to the first lock (40) and the second lock (50).

(53) Although the present disclosure has been described in detail with reference to a specific embodiment, it is apparent to those skilled in the art that various changes and modifications can be made without departing from the spirit and scope of the present disclosure.

Claims

1. A connector comprising: a housing having a first connection portion and a second connection portion adjacent to the first connection portion; a first cap pivotally supported by the housing to be displaceable between an open position where the first connection portion is exposed to outside and a closed position where the first connection portion is shielded from the outside; a second cap pivotally supported by the housing to be displaceable between an open position where the second connection portion is exposed to the outside and a closed position where the second connection portion is shielded from the outside; a first lock rotatably supported by the housing to lock the first cap when the first cap is located at the closed position; a second lock rotatably supported by the housing to lock the second cap when the second cap is located at the closed position; and a power transmission member rotatably supported by the housing, wherein the power transmission member is configured to transmit a rotational force in a lock releasing direction of the second lock to the first lock as a rotational force in a lock releasing direction of the first lock such that the first cap and the second cap are integrally displaced to the open position.

2. The connector according to claim 1, wherein the power transmission member includes a first contact portion to come into contact with the first lock only on a locking direction side, and a second contact portion to come into contact with the second lock only on a lock releasing direction side.

3. The connector according to claim 1, wherein the power transmission member is supported by a rotation shaft common to the first lock and the second lock.
